MIL-T-19147D
16 November 1984
SUPERSEDING
MIL-T-19147C
18 June 1976

MILITARY SPECIFICATION

TRUCKS, DOLLY, RECTANGULAR, WITH 4 SWIVEL CASTERS

(GENERAL PURPOSE AND MACHINERY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rectangular frame, general purpose, and machinery dolly trucks, each with four swivel casters.

1.2 Classification. Dolly trucks shall be of the following types, as specified (see 6.2):

Type I - General purpose, 1000-pound capacity, with rubber "mold-on" wheels.
Type II - Machinery, 2400-pound capacity, with steel or plastic wheels.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3920

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MS24380	- Casters, Industrial.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.
D3951 - Standard Practices for Commercial Packaging.
D2016 - Moisture Content of Wood, Test Methods for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The dolly trucks shall consist of a rectangular steel frame and four swivel casters as shown in figures 1 and 2. The type II truck shall have two platform boards attached to the frame.

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3.2 First article . Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of trucks from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Steel. Unless otherwise specified (see 6.2), steel shall conform to ASTM A36.

3.3.2 Wood. The wood for the two platform boards shall be any one of the following species: Ash (white), Beech, Birch, Hickory, Maple (hard), Oak, Pecan, or Rock Elm.

3.3.2.1 Quality. The boards shall be free from decay, holes, loose knots, larvae channels, shake, split, warp, brashness, and slope of grain exceeding 1 inch in 10 inches. Sound, tight knots that do not exceed one-third the width of the board in which they appear, and stain which does not obscure the grain, will be permitted.

3.3.2.2 Moisture content. At the time of fabrication of the truck, the moisture content of the lumber for the wood boards shall be not less than 12 percent nor more than 19 percent.

3.4 Design and construction.

3.4.1 Frame. The truck frame shall be fabricated of steel as specified in 3.3.1, and shall be of all welded construction.

3.4.1.1 Capacities and sizes. The capacities and sizes of the trucks shall be as shown in table I.

TABLE I. Capacities and sizes.

Type	Width and length (inches, ± 1/2 inch)	Max, height to top of frame (inches)	Rated capacity (lbs., min.)
I	24 x 30	8	1000
II	18 x 24	8	2400

3.4.2 Casters. The casters shall be securely bolted to the underside of the frame, each located as close as practical to each of the four corners. If corners are rounded, the caster mounting plate shall be rounded concentrically or flush with the frame. Bolts shall not protrude above the top surface of the frame. Casters shall be in accordance with MS24380, part No. 6SU for type I truck and part No. 6SC or 6SM for type II truck.

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3.4.3 Platform. The platform boards shall be fabricated of any one of the woods specified in 3.3.2, and shall conform to the quality and the moisture content specified in 3.3.2.1 and 3.3.2.2, respectively. Each board shall be not less than 13/16 inch thick and 7-1/2 inches wide. Length shall be as required for the ends to fit flush with the 24-inch sides of the frame. Except for the bottom surface, all surfaces of each board shall be smoothly finished, and a 1/8-inch x 45-degree chamfer shall be placed on all top edges. The two boards shall be securely bolted with not less than eight bolts (four bolts on each board) to the top of the frame so that one side of each board is flush with each of the outer 18-inch sides. Bolts shall be not less than 3/8 inch in diameter, and bolt heads shall not protrude above the board surface. When mounted, boards shall be supported by the frame on all four sides and shall not extend beyond the frame on any side. The inner sides (along the board length) may be inset from the inner frame supports not more than 1/8 inch.

3.4.4 Fastening devices. All bolts, capscrews, and nuts shall be provided with lockwashers or other locking devices and shall be tightly drawn. All hardware shall be treated and plated in accordance with commercial practice to prevent corrosion.

3.5 Performance.

3.5.1 Static load. The truck shall withstand static load of 200 percent of rated capacity (see table I) without evidence of damage, weld failure, distortion, or permanent deformation when tested in accordance with 4.5.2.2.

3.5.2 Rolling resistance. The truck shall be capable of being pushed or towed with a maximum force no greater than 2-1/2 percent of rated load plus the weight of the truck.

3.6 Identification marking. Each truck shall be identified with an identification plate containing the following information: Ownership designation, nomenclature, contract number, national stock number (NSN), and manufacture. The plate shall conform to MIL-P-514, type III, composition C (GG-5-455) type I, grade A, class D unless otherwise specified. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be securely attached to the truck.

3.7 Treatment and painting. The parts of the truck normally painted shall be cleaned, treated and painted in accordance with the contractor's commercial practice.

3.8 Workmanship. The trucks shall be clean and correctly assembled. All burrs, splinters, and sharp edges shall be removed. No part shall be impaired or damaged in any way or show any evidence of rust or corrosion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract,

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the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be subjected to the tests as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each truck shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The truck shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Wood not as specified.
103. Design not as specified.
104. Construction not as specified.
105. Dimensions not as specified.
106. Fasteners not as specified.
107. Workmanship not as specified.
108. Identification marking not as specified.
109. Treatment and painting not as specified.

4.5.2 Tests.

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4.5.2.1 Moisture content. Determination of the moisture content of the wood shall be made in accordance with ASTM D2016, electric moisture meter method. Three readings shall be taken. Nonconformance to 3.3.2.2 shall constitute failure of this test.

4.5.2.2 Static load. The truck shall be placed on a dry, smooth, level, concrete floor, and loaded with a uniformly distributed load equal to 200 percent of rated capacity as shown in table I. The truck shall support this load for a period of 1 hour after which the load shall be removed and the truck examined. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.3 Rolling resistance. The truck, uniformly loaded up to its rated capacity, shall be weighed and then placed on a dry, smooth, level, concrete surface. The truck with rated load shall be accelerated to a minimum speed of 2 miles per hour (MPH) by means of a suitable towing device employing a tension dynamometer. The maximum gage reading of the dynamometer at this speed shall be recorded over a minimum travel distance of 25 feet. This test shall be repeated in the opposite direction so as to traverse the initial path and terminate at the starting point. Nonconformance to 3.5.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

110.	Trucks not nested as specified for level A.
111.	Dunnage not provided and placed as specified for level A.
112.	Strapping not as specified for level A.
113.	Strapping not applied as specified for level A.
114.	Metal edge protectors not applied as specified for level A.
115.	Marking missing, illegible, incorrect, or incomplete for level A, or commercial.
116.	Packaging not as specified.

5. PACKAGING

5.1 Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Two trucks shall be nested in the most compact manner: wheels to wheels. Dunnage lumber shall be used to provide a 3/4-inch minimum clearance between the tread of the wheels and the undersurface of the opposing truck and to immobilize the casters. The nested trucks shall be strapped

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together with three flat steel straps, not less than 3/4 inch wide x 0.023 inch thick, conforming to QQ-S-781, class 1, type I or IV. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A. Two straps shall be positioned widthwise, equally spaced and one strap shall be centered lengthwise. Metal edge protectors shall be provided where strapping passes over the edge of the trucks.

5.1.2 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Commercial packs shall be marked in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. Type I trucks are intended for moving loads (not exceeding 1000 pounds) over short distances on paved surfaces. Type II trucks are intended for use singly or in pairs for moving machinery and other bulky loads (not exceeding 2400 pounds per truck) over short distances on paved surfaces.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. When first article is not required (see 3.2).
d. Time frame required for submission of first-article model and number of trucks required (see 3.2).
e. Type of steel required if other than as specified (see 3.3.1).
f. Degree of packing required (see 5.1).
g. When strapping shall be other than as specified (see 5.1.1).

6.3 First-article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units of each type specified (see 6.2). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

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Army - ME

Review activities:
Army - AT
DLA - GS

Project 3920-0158

User activity:
Navy - MC

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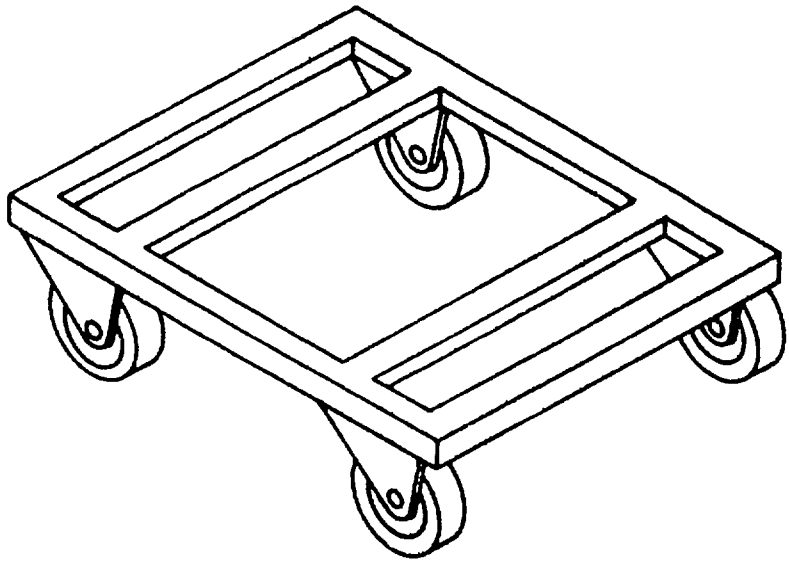


FIGURE I. GENERAL PURPOSE DOLLY-TYPE I

CX-1366

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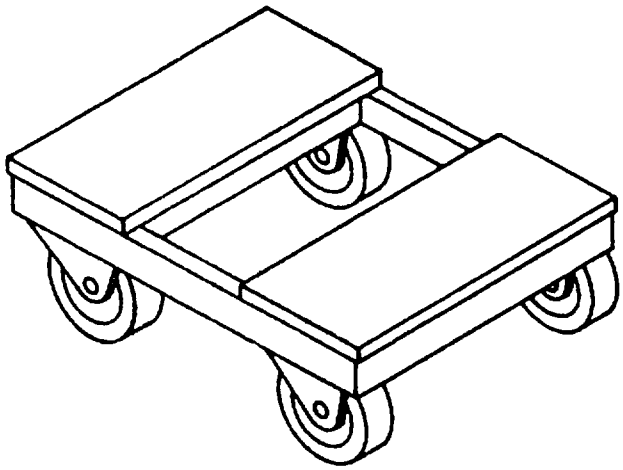


FIGURE 2. MACHINERY DOLLY - TYPE II

CX-1367

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-19147D	Trucks, Dolly, Rectangular, with 4 Swivel Casters (General Purpose and Machinery)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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